Breast Cancer AI — Four Individual Project Proposals

Hospital-Partnered Capstone | Individual Pitch: 10 minutes, 4 slides

Required slide structure (per course announcement):

1. Problem Statement
2. Data Availability
3. State of the Art (SOTA) / Relevant Work
4. Proposed Method & Expected Outcome

Course requirement to satisfy on Slide 2: *"If you select a project from the hospital-provided list, you must also identify publicly available data that can be used to address the same problem, in case the hospital is unable to provide the required dataset."*

Strategy Summary (for the 4 of you — not for the slides)

#	Proposal	Base Hospital Project	Partner	Core Task	Primary Metric
A1	Pretreatment Response Prediction	MRI Based Assessment of Chemo Response	MNJ	Binary classification	AUC
A2	Post-Treatment Residual Disease Assessment	MRI Based Assessment of Chemo Response	MNJ	Segmentation → classification	Dice, then AUC
B1	Pathology + Genomics Molecular Subtyping	Multimodal Breast Cancer Diagnostic & Outcome Assistant	Kamineni	Multi-label classification	AUC / accuracy
B2	Graph-Based Prognosis with Missing-Modality Robustness	Multimodal Breast Cancer Diagnostic & Outcome Assistant	Kamineni	Survival analysis	C-index

Why these two base projects: both have fully annotated, ready-to-use public data (no manual annotation required by anyone on the team); both split naturally into two approaches; they come from two different hospital partners; and all four share breast cancer as the disease, so the later team merge is coherent.

Merge story (Section at the end of this document): the four proposals form one continuous clinical pathway — diagnosis (B1) → treatment planning (A1) → treatment assessment (A2) → outcome (B2).

Compute available: NVIDIA RTX 6000 (Turing, 24 GB) + IIIT Ada HPC cluster (SLURM; 4× GTX 1080 Ti / RTX 2080 Ti, 11 GB each; PyTorch 2.4.1+cu118, Python 3.12).

> **Technical note:** Turing supports **fp16 tensor cores but NOT bf16** (bf16 is Ampere and later). Use `torch.cuda.amp.autocast(dtype=torch.float16)` with a `GradScaler`. 24 GB is enough to train full 3D DCE-MRI volumes without patch-based workarounds.

PROPOSAL A1

Pretreatment MRI-Based Prediction of Response to Neoadjuvant Chemotherapy

Base project: MNJ — *MRI Based Assessment of Chemo Response*

SLIDE 1 — Problem Statement

Breast cancer patients receive neoadjuvant chemotherapy (NAC) before surgery. An MRI is acquired at the start of chemotherapy and another after it completes. Surgery follows, and the pathology report definitively establishes the presence or absence of residual disease.

Roughly **one third of patients achieve pathologic complete response (pCR)** — in the I-SPY2 trial cohort, 217 of 660 patients (32.88%). The remaining two thirds undergo months of cytotoxic therapy with significant toxicity and no complete response.

The clinical question: *Can we predict, from the pretreatment MRI alone, which patients will achieve pCR?*

Why it matters: A reliable pretreatment predictor supports treatment selection at the point where the decision is still reversible — escalating or switching regimens for predicted non-responders, rather than discovering failure after the full course. This is upstream of, and complementary to, the post-treatment assessment question.

Scope of this project: binary pCR prediction from pretreatment imaging plus clinical/receptor variables. Explicitly *not* in scope: treatment recommendation, or any autonomous clinical decision.

SLIDE 2 — Data Availability

Hospital data (MNJ):

- MRI images pre-chemotherapy with pathological confirmation
- MRI images post-chemotherapy, with and without residual disease, pathologically confirmed
- Status per project sheet: ~100 CDs; requires IRB approval; ethics approval in progress
- Reference DOI supplied by MNJ: https://doi.org/10.4238/n53tzp26

Public fallback — fully labelled, zero annotation required:

Primary: BreastDCEDL_ISPY2 — n = 982 pre-treatment DCE-MRI cases

- Zenodo: https://zenodo.org/records/17578255
- Contents: DCE-MRI sequences (3–12 timepoints per scan, typically 7), derived maps, full 3D tumor segmentations
- Labels already present: **pCR**, hormone receptor (HR) status, HER2 status, MammaPrint risk level, age at screening
- Demo notebook: https://github.com/naomifridman/BreastDCEDL/blob/main/ISPY2/BrestDCEDL_ISPY2_demo.ipynb

Parent dataset: BreastDCEDL — 2,070 patients (I-SPY1 + I-SPY2 + Duke)

- Paper (*Scientific Data*, 2026): https://www.nature.com/articles/s41597-026-06589-6

- Zenodo: https://zenodo.org/records/15627233
- GitHub: https://github.com/naomifridman/BreastDCEDL
- Unified tumor annotations (binary masks for I-SPY cohorts, bounding boxes for Duke); 49-variable harmonized clinical metadata; **pCR available for 1,452 patients — 100% of I-SPY1 and I-SPY2**
- License: CC BY-NC 4.0 (I-SPY1 source CC BY 3.0; I-SPY2 source CC BY 4.0)

Original TCIA collections (if you prefer raw DICOM):

- I-SPY2 / ACRIN 6698: https://doi.org/10.7937/TCIA.D8Z0-9T85 — wiki: https://wiki.cancerimagingarchive.net/pages/viewpage.action?pageId=70230072
- ACRIN 6698 / I-SPY2 Breast DWI: https://doi.org/10.7937/TCIA.KK02-6D95
- I-SPY1 / ACRIN 6657: https://wiki.cancerimagingarchive.net/pages/viewpage.action?pageId=20643859
- Duke Breast Cancer MRI: https://doi.org/10.7937/TCIA.e3sv-re93
- TCIA portal: https://www.cancerimagingarchive.net

Key point for the slide: TCIA and Zenodo require no credentialing, no CITI training, and no institutional data-use agreement. Data is downloadable immediately, already segmented and already labelled by the trials themselves.

SLIDE 3 — State of the Art / Relevant Work

Work	Data	Result
Attention-based multimodal pCR prediction (2025)	I-SPY2 (n=660) train, I-SPY1 (n=114) external	**AUC 0.73 ± 0.04 internal, 0.71 ± 0.02 external**. MRI-only: 0.68/0.70. Clinical-only: 0.66/0.71
Pretreatment TNBC model + clinicopathologic data	282 patients internal, I-SPY2 external	**AUC 0.76 internal, 0.72 external**. Tumor-volume preprocessing mattered; 3D framework choice and ROI selection had minimal impact
Two-stage dual-task early prediction	I-SPY2, T0/T1/T2 timepoints	Single-stage single-task **0.799** → two-stage dual-task **0.820**
Integrated CNN (longitudinal multiparametric MRI + demographics + molecular subtype)	I-SPY1 (N=155 after exclusions, 9 institutions)	Predicts pCR, RCB and PFS; outperformed stacking and concatenation fusion
DL-based pharmacokinetic quantification of early-treatment DCE-MRI	ACRIN 6698 / BMMR2 challenge (A1 train, A2 test), I-SPY2, I-SPY1	Radiomics on PK maps + clinicopathologic variables → logistic regression for pCR

Three observations to state explicitly on the slide:

1. **Performance is plateaued in the 0.70–0.82 range.** This field is *not* saturated, unlike ECG classification or lung-nodule detection. There is genuine headroom.

2. **Multimodal fusion beats either modality alone, consistently.** MRI-only ≈ 0.68, clinical-only ≈ 0.66, fusion ≈ 0.73. This is a reliable, reproducible finding you can build a planned experiment around.
3. **Cross-cohort generalization degrades.** Every study above reports an external-validation drop. Any honest proposal must plan for it.

SLIDE 4 — Proposed Method & Expected Outcome

Pipeline:

```
Pretreatment DCE-MRI volume (3D, multi-timepoint)
        |
        ├─▶ 3D CNN encoder (3D ResNet-18 / ResNet-50, or Video-ViT)
        |        └─ tumor-region crop guided by provided segmentation masks
        |
Clinical/receptor vector (HR, HER2, MammaPrint, age)
        |
        └─▶ MLP encoder
                  |
            Cross-attention fusion
                  |
            Binary head → P(pCR)
```

Experimental plan (deliberately structured as an ablation, so the result is informative either way):

1. **Baseline 1:** clinical features only (logistic regression / gradient boosting) — target ≈ 0.66
2. **Baseline 2:** imaging only (3D CNN) — target ≈ 0.68
3. **Proposed:** cross-attention fusion — target ⩾ 0.73
4. **External validation:** train on I-SPY2, test on I-SPY1 — report the drop honestly

Validation protocol: patient-level stratified splits (never slice-level), stratified by pCR status and HR/HER2 subtype. 5-fold cross-validation. Report AUC with confidence intervals, plus sensitivity/specificity at a clinically chosen operating point.

Interpretability: 3D Grad-CAM over the tumor volume; attention weights over clinical features.

Expected outcome:

- A reproducible pCR prediction pipeline with AUC in the 0.70–0.78 range
- A quantified answer to "how much does each modality contribute?"
- An honest external-validation number showing the I-SPY2 → I-SPY1 generalization gap
- If MNJ data arrives: a second external validation on an Indian cohort, which would be the most interesting result in the project

Risks and mitigations:

- *Class imbalance* (~33% pCR) → stratified sampling, class-weighted loss, report AUPRC alongside AUC
- *Small effective N* → transfer learning from Kinetics/RadImageNet pretrained 3D backbones; heavy augmentation

- *MNJ data may not arrive* → the entire project runs on public data; hospital data is a bonus external validation set, not a dependency

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PROPOSAL A2

Post-Chemotherapy Residual Disease Assessment from Breast MRI

Base project: MNJ — *MRI Based Assessment of Chemo Response*

SLIDE 1 — Problem Statement

This is MNJ's problem statement, almost verbatim:

> Patients have an MRI at the start of chemotherapy and one following chemotherapy. Patients also undergo surgery after chemotherapy and the pathology report proves the presence or absence of any residual disease. **If the response to the chemotherapy can be assessed through an MRI, we can avoid the subsequent surgery if the response to chemotherapy is complete (absence of any residual disease).**

The problem: Surgery is currently the only way to *confirm* pathologic complete response. But surgery is also the thing you would want to avoid if pCR has already been achieved. This is a circular dependency that only imaging can break.

The clinical question: *Can AI algorithms establish the presence or absence of residual cancer from post-chemotherapy MRI, with sufficient reliability to inform the surgical decision?*

Framing note for the pitch: the bar here is asymmetric. A false negative (declaring pCR when residual disease remains) means a missed cancer. The operating point must be chosen for **high sensitivity to residual disease**, and the proposal should say so explicitly. This asymmetry is the most interesting design constraint in the project and distinguishes it from a generic classification task.

SLIDE 2 — Data Availability

Hospital data (MNJ): as A1 — pre- and post-chemotherapy MRI with pathological confirmation of residual disease. ~100 CDs, IRB pending. https://doi.org/10.4238/n53tzp26

Public fallback — segmentation masks already expert-verified:

Primary: MAMA-MIA — 1,506 DCE-MRI cases with expert segmentations

- Paper (*Scientific Data*, 2025): https://www.nature.com/articles/s41597-025-04707-4
- Sources: I-SPY1/ACRIN 6657, I-SPY2/ACRIN 6698, NACT-Pilot, Duke-Breast-Cancer-MRI (all via TCIA)
- **Segmentations were produced automatically then corrected and verified by 16 breast cancer experts averaging 9 years of experience** — a fully annotated dataset
- Includes 49 harmonized variables (21 clinical, 6 demographic, 22 imaging parameters)
- **Ships with pretrained nnU-Net baseline weights** — you start from a working segmentation model, not from scratch

Secondary: BreastDCEDL I-SPY1 subset

- https://zenodo.org/records/15627233 — the I-SPY1 subset provides **3–5 DCE scans per patient with complete tumor segmentations**, explicitly described as suited to temporal analysis

- I-SPY1 imaging timepoints: T1 (pre-AC), T2 (after cycle 1 AC), T3 (between AC and taxane), T4 (post-chemo, pre-surgery) — https://wiki.cancerimagingarchive.net/pages/viewpage.action?pageId=20643859

I-SPY2 FTV Analysis Masks (native TCIA): bit-encoded segmentations encoding all masking steps from the I-SPY2 primary functional tumor volume analysis, with SER derived objects. Documentation files ("Analysis mask files description", "ACRIN 6698 ISPY2 Shared Private Tag Data Dictionary", "ACRIN 6698 ISPY2 DWI and DCE MRI Data Descriptions") are downloadable from the collection page: https://wiki.cancerimagingarchive.net/pages/viewpage.action?pageId=70230072

> Caveat stated on the TCIA page: FTV results in the SER derived objects are **not identical** to those used in the I-SPY2 study, due to differing software implementations. Mention this if you use FTV as a baseline.

> **Action item before finalizing:** confirm how many *post-chemotherapy* (T4) timepoints carry segmentation masks. If post-chemo masks are sparse, pivot the task to **longitudinal change modelling (T0 → T1 → T2)**, which has a published benchmark (0.799 → 0.820) and keeps this proposal distinct from A1.

SLIDE 3 — State of the Art / Relevant Work

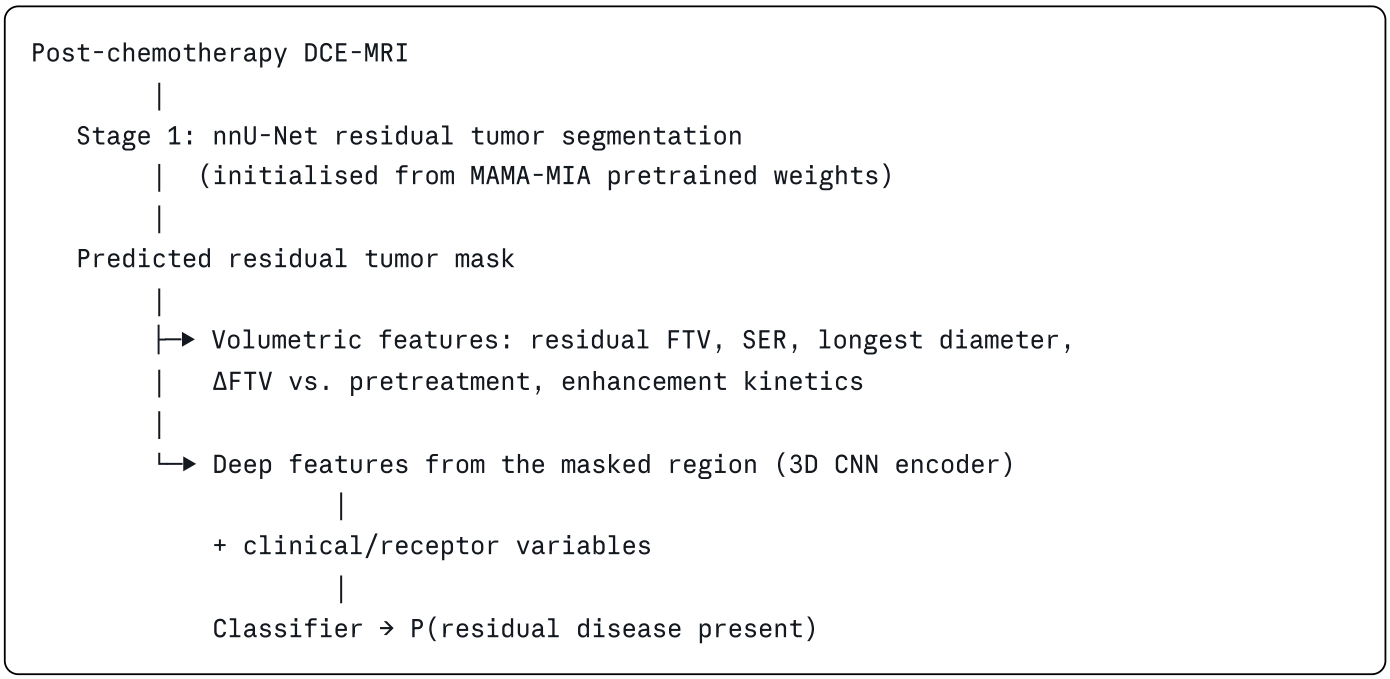
Work	Relevance
MAMA-MIA benchmark (2025)	Establishes the segmentation benchmark and provides pretrained nnU-Net weights; adopts a BraTS-style automatic-then-expert-corrected annotation strategy
Two-stage dual-task learning (I-SPY2, T0/T1/T2)	Trains ConvLSTM on all timepoints to predict pCR and extract the T2 latent representation, then a dual-task model predicts both pCR and the T2 representation from T0+T1 only. AUROC 0.799 → **0.820**. Motivation: relying on mid/post-treatment images outweighs the benefit of adaptive treatment
Integrated CNN, longitudinal multiparametric MRI (I-SPY1, N=155, 4 timepoints)	Predicts pCR, RCB and PFS. Code: https://github.com/HongyiDuanmu26/Prediction-of-pCR-with-Integrative-Deep-Learning — paper: https://journals.plos.org/plosone/article?id=10.1371%2Fjournal.pone.0280148
Functional tumor volume literature (Hylton et al.)	Strong correlation between pCR and DCE-derived functional tumor volume; FTV change useful for recurrence-free survival — the classical non-DL baseline you should beat
CNN on I-SPY TRIAL breast MRI (improved InceptionV3)	Precision 0.83, recall 0.77, F1 0.80, accuracy 0.81, AUC 0.81 on the test split

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Framing for the slide: segmentation-based and volumetric approaches (FTV) represent the clinically accepted method; deep learning has matched but not decisively surpassed them. A segmentation-first pipeline lets you compare directly against the clinical standard rather than against another black box.

SLIDE 4 — Proposed Method & Expected Outcome

Pipeline — segmentation-first, deliberately different from A1's end-to-end classification:



Experimental plan:

1. **Stage 1 evaluation:** Dice / IoU against expert masks. Compare fine-tuned vs. off-the-shelf MAMA-MIA weights
2. **Baseline:** classical FTV threshold (the clinical standard) — the number to beat
3. **Proposed:** segmentation-derived features + deep features + clinical → residual disease classification
4. **Operating point analysis:** ROC curve with explicit selection of a **high-sensitivity threshold**, reporting the specificity cost of guaranteeing e.g. 95% sensitivity to residual disease

Validation: patient-level splits; 5-fold CV; external validation across cohorts (train on I-SPY2, test on I-SPY1, or vice versa).

Interpretability: overlay predicted residual masks on the post-chemo volume next to the expert mask. This is the most visually compelling slide in any of the four proposals — use it.

Expected outcome:

- Residual tumor segmentation with Dice competitive with the MAMA-MIA baseline
- A residual-disease classifier benchmarked against the FTV clinical standard
- **A sensitivity/specificity trade-off curve framed around the surgical decision** — i.e. "at the threshold where we miss no residual disease, what fraction of patients could still safely avoid surgery?" This is the number MNJ actually cares about
- A clear statement of what performance would be required before such a tool could responsibly inform surgical planning (and an acknowledgement that this project does not reach that bar)

Risks and mitigations:

- *Sparse post-chemo masks* → pivot to longitudinal T0→T1→T2 change modelling (published benchmark exists)

- *Segmentation errors propagate to classification* → report end-to-end performance with both predicted and ground-truth masks, to isolate the error source
- *Small residual-disease-positive subgroup in some subtypes* → stratify by HR/HER2 status, report per-subtype performance

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PROPOSAL B1

Multimodal Fusion of Histopathology and Genomics for Breast Cancer Molecular Subtyping

Base project: Kamineni — *Multimodal Breast Cancer Diagnostic & Outcome Assistant*

SLIDE 1 — Problem Statement

From Kamineni's project brief:

> Breast cancer information is distributed across pathology, radiology, clinical history, biopsy, treatment and follow-up records. [...] Develop a multimodal AI system integrating pathology, radiology and clinical data to support diagnosis, risk stratification and outcome prediction. **Workflow fit:** Pathology/Radiology → Clinical review/MDT → Treatment planning → Follow-up. AI provides a consolidated case view and highlights relevant findings for clinician review.

The specific problem this proposal addresses: Molecular subtype (PAM50) and receptor status (ER, PR, HER2) drive treatment selection, but establishing them requires IHC and/or molecular assays that are expensive, slow, and unevenly available. Meanwhile, a routine H&E slide already exists for every patient.

The clinical question: *How much molecular information is recoverable from routine histopathology, and how much does adding transcriptomics improve it?*

Scope: predicting ER, PR, HER2 and PAM50 subtype from whole-slide images and RNA-seq. Explicitly not in scope: replacing IHC, or any autonomous diagnosis.

SLIDE 2 — Data Availability

Hospital data (Kamineni): Status is "Awaiting" per the project sheet. Kamineni's brief states they already have ongoing AI research across pathology, breast cancer, radiology and outcomes.

Public fallback — labels are pre-existing molecular ground truth, no annotation required:

Primary: TCGA-BRCA (~1,000 patients)

- GDC portal: https://portal.gdc.cancer.gov/projects/TCGA-BRCA
- Imaging (diagnostic + tissue slides) via TCIA: https://www.cancerimagingarchive.net
- Modalities: H&E whole-slide images, RNA-seq, CNV, DNA methylation, miRNA, clinical variables
- Labels already present: **PAM50 subtype, ER status, PR status, HER2 status, survival** — derived from molecular assays performed by the TCGA consortium, not by annotators
- **Critical point for the slide:** WSI is used with **slide-level labels under multiple-instance learning**. No pixel or patch annotation is needed at any stage.

External validation: CPTAC-BRCA (~134 patients) — adds proteomics, enabling a genuine cross-cohort test.

Preprocessing tooling (all open source, all standard):

- Stain normalization: Macenko et al., *A method for normalizing histology slides for quantitative analysis*, ISBI 2009
- WSI handling / tiling: QuPath — Bankhead et al., *Sci Rep* 7:16878 (2017); OpenSlide
- Batch correction for omics: ComBat
- RNA-seq quantification: RSEM — Li & Dewey, *BMC Bioinformatics* 12:323 (2011)

Access note: TCGA open-tier data (WSI, expression, clinical) requires no credentialing. Only controlled-access raw sequence data requires dbGaP authorization, which this project does not need.

SLIDE 3 — State of the Art / Relevant Work

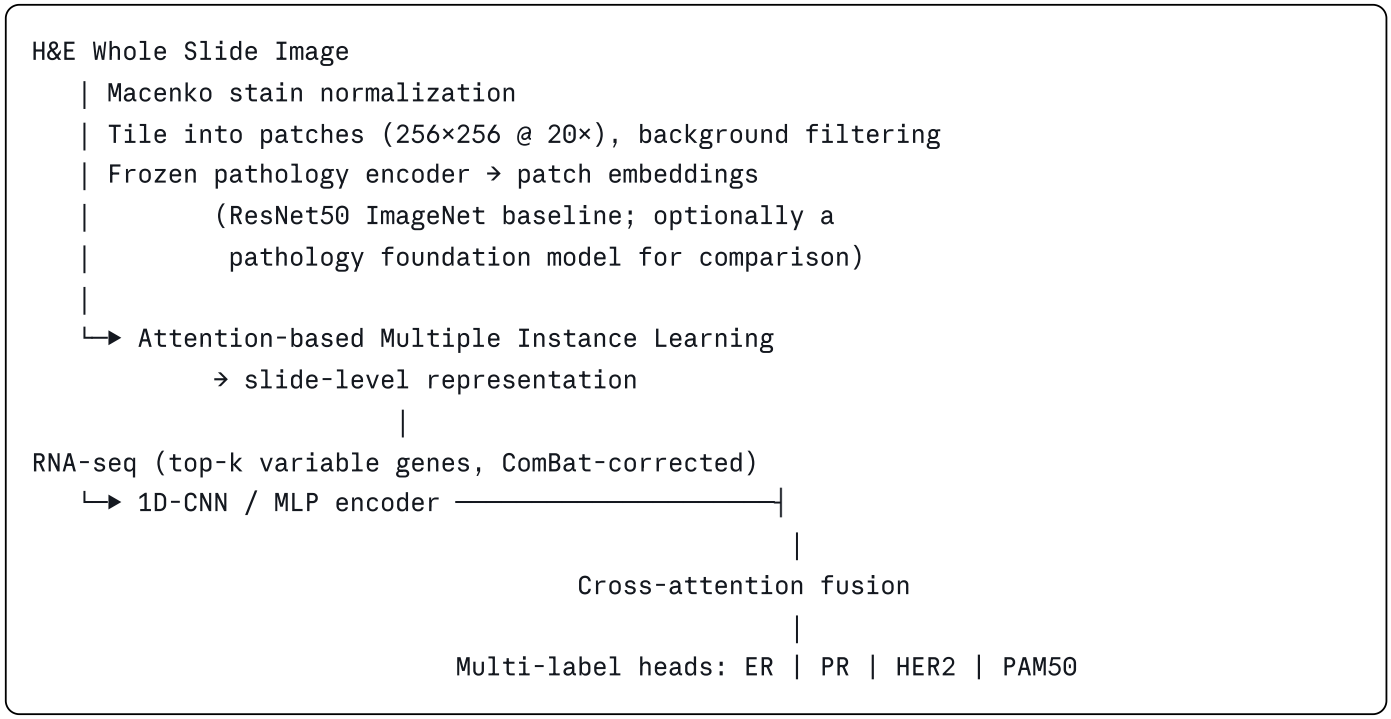
Work	Data	Result
M3FusionNet (Multi-modal, Multi-scale, Multi-source)	TCGA-BRCA n≈1036, CPTAC-BRCA n≈134	H&E WSI (ResNet50, 2048-d embeddings) + RNA-seq + miRNA + proteomics + clinical, feature-level fusion, XGBoost/CatBoost heads. **In-distribution AUC: 0.99 ER, 0.96 PR, 0.98 HER2, 0.96 PAM50 macro-AUC; overall 0.97.** Cross-dataset drop of **13–22%** (overall 0.82). Combined TCGA+CPTAC model: overall 0.90. Proteomics improved HER2 AUC by 2–3 points. Used Macenko normalization + ComBat
Selective/routing multimodal subtype classification	TCGA-BRCA, 924 patients with PAM50	Uncertainty-aware routing: RNA-only for high-confidence cases, multimodal for ambiguous ones. **94.93% accuracy**, beating both unimodal baselines and fixed fusion. Multimodal inference needed for only **38.2%** of samples → 3.12× speedup. Notable gains distinguishing Luminal A from Luminal B
HEALNet	Multimodal heterogeneous biomedical data	Cross-attention fusion architecture; NeurIPS 2024
DeepClinMed-PGM (Wang et al., 2024)	TCGA + SYSMH cohorts	Pathology imaging + genomic + clinical, hierarchical fusion, AUC 0.979, Grad-CAM interpretability
Cross-attention mammography + clinical with explainability	—	https://arxiv.org/pdf/2508.16000 — includes a useful comparison table of 2023–2025 multimodal breast cancer models
Explainable multi-modal DL (radiology + clinical)	CBIS-DDSM, Duke Breast MRI, TCGA (BRCA/LUAD/GBM) — 3,842 images, 2,917 patients	Attention-based fusion outperformed feature concatenation; Grad-CAM++ for images, SHAP for clinical features; explainability did not cost accuracy

The open problem to name on the slide: in-distribution performance is near-ceiling (0.96–0.99), but **cross-cohort transfer loses 13–22% AUC.** The interesting question is no longer "can we fuse?" but

"does the fusion survive contact with a different cohort?"

SLIDE 4 — Proposed Method & Expected Outcome

Pipeline:



Experimental plan:

1. **Baseline 1:** RNA-seq only (the known-strong modality) — this is the honest bar
2. **Baseline 2:** WSI only (attention-MIL)
3. **Baseline 3:** naive concatenation fusion
4. **Proposed:** cross-attention fusion
5. **Generalization test:** train on TCGA, evaluate on CPTAC — quantify the drop and compare against the reported 13–22%

Key design decision to defend in the pitch: RNA-seq alone is already very strong for subtyping. The proposal must therefore be explicit that **the contribution is not "beat RNA-seq" but "recover molecular information from the slide that is already in the pathologist's hands"** — i.e. how much of the molecular signal is visible in tissue morphology, and where. That reframing is what makes this a research question rather than a leaderboard exercise.

Interpretability: attention heatmaps over WSI tiles showing which tissue regions drive each receptor prediction; SHAP over gene features.

Expected outcome:

- Multi-label classifier with per-biomarker AUCs benchmarked against published TCGA numbers
- A quantified modality-contribution analysis (how far does morphology alone get you?)
- A cross-cohort generalization number on CPTAC
- Attention heatmaps linking predicted receptor status to specific morphological regions

Risks and mitigations:

- *WSI storage and compute* → 24 GB RTX 6000 handles MIL over pre-extracted patch features comfortably; precompute and cache embeddings once, then train the fusion head repeatedly (this is the standard workflow and keeps iteration fast)
- *Class imbalance in PAM50* (HER2-enriched and Normal-like are rare) → macro-averaged metrics, class weighting, report per-class results
- *Batch effects between TCGA sites* → Macenko normalization + ComBat, as in M3FusionNet

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PROPOSAL B2

Graph-Based Multimodal Prognostic Modelling with Missing-Modality Robustness

Base project: Kamineni — *Multimodal Breast Cancer Diagnostic & Outcome Assistant*

SLIDE 1 — Problem Statement

Kamineni's brief asks for a system supporting *"diagnosis, risk stratification and outcome prediction"* with *"potential for personalized risk/outcome prediction."* This proposal takes the **outcome** half.

Two problems, addressed together:

(1) Prognosis is not classification. Predicting *whether* a patient has a given subtype and predicting *when* an event will occur are different statistical problems. Survival data is censored — most patients have not had the event at last follow-up — so accuracy and AUC are the wrong instruments. This requires time-to-event modelling and a C-index, not a confusion matrix.

(2) Complete modalities are a laboratory fiction. In real clinical practice, genomic sequencing or high-resolution imaging is frequently missing. Models trained assuming full modality availability degrade severely when a modality is absent. Kamineni's own workflow description — pathology and radiology feeding an MDT — implies exactly this heterogeneity.

The research question: *Can a graph-structured fusion of tissue architecture and molecular pathways stratify survival risk, and does it degrade gracefully when modalities are missing or the cohort changes?*

SLIDE 2 — Data Availability

Hospital data (Kamineni): "Awaiting." Their brief notes existing internal AI work on breast cancer and outcomes.

Public fallback:

Development: TCGA-BRCA — https://portal.gdc.cancer.gov/projects/TCGA-BRCA

- WSI + RNA-seq + CNV + methylation + miRNA + clinical
- **Survival labels (overall survival, progression-free interval, censoring indicators) are part of the standard TCGA clinical release** — no annotation required
- Slides via TCIA: https://www.cancerimagingarchive.net

External validation: CPTAC-BRCA (~134 patients, adds proteomics) — enables a genuine cross-cohort survival test, which most published work omits.

Optional radiology arm: TCGA-BRCA has a TCIA imaging collection (Lingle et al., *The Cancer Genome Atlas Breast Invasive Carcinoma Collection (TCGA-BRCA)*, The Cancer Imaging Archive,

2016), allowing a radiology–pathology–genomics triple if time permits.

Biological structure for the graph (no annotation, publicly available):

- Pathway databases — Reactome, KEGG, MSigDB Hallmark gene sets — to define the omics graph topology
- Tissue graph built from WSI patch coordinates (spatial k-NN over tiles), so the graph is derived, not annotated

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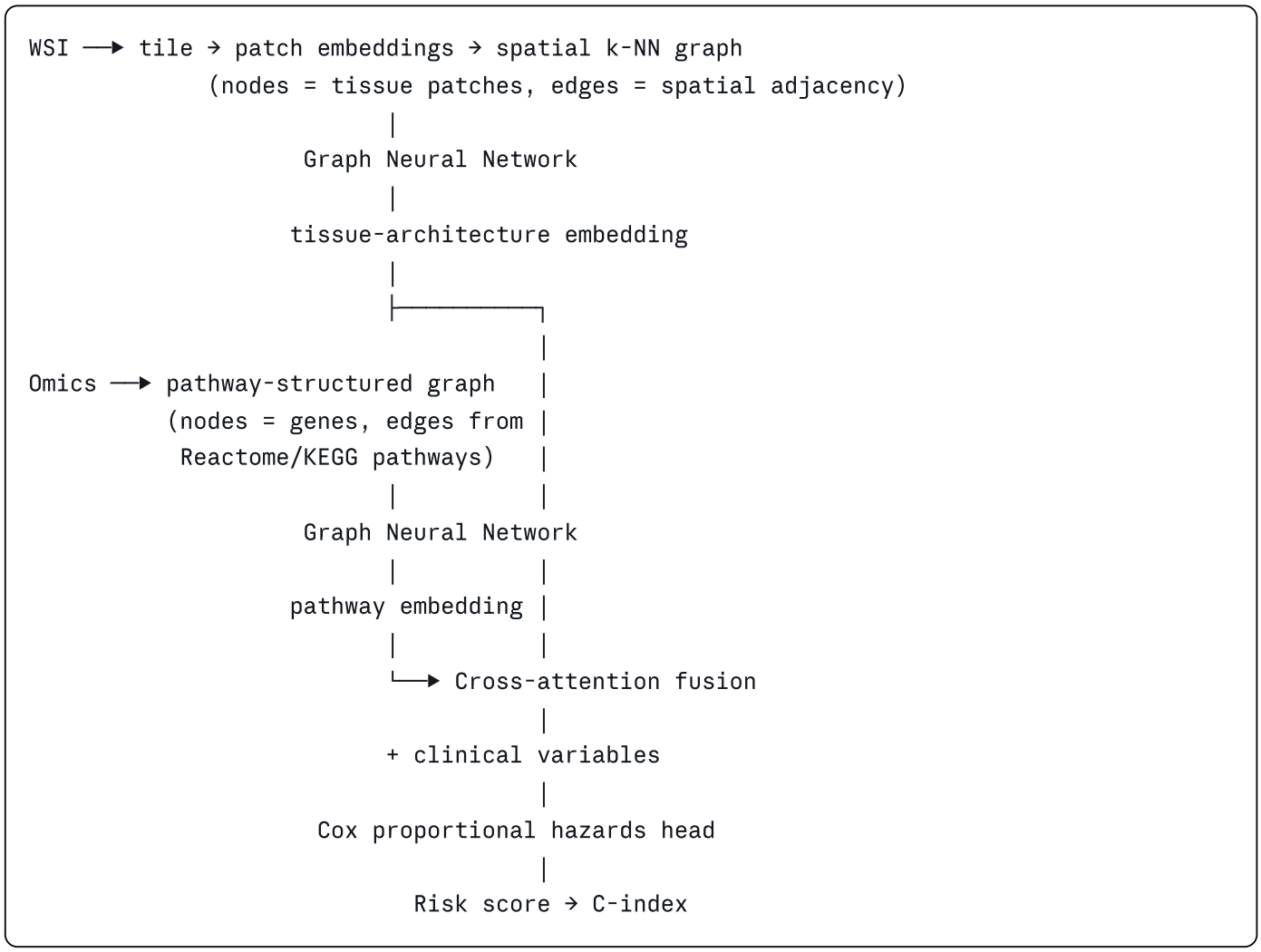
SLIDE 3 — State of the Art / Relevant Work

Work	Relevance
MM-SurvNet — Deep learning-based survival risk stratification in breast cancer through multimodal data fusion	Directly the same task. https://arxiv.org/pdf/2402.11788
M3FusionNet (TCGA n≈1036, CPTAC n≈134)	Reports the central failure mode: cross-dataset generalization drops **13–22% AUC**, mitigated by harmonized preprocessing (Macenko + ComBat) and multimodal fusion. Combined TCGA+CPTAC training reached overall AUC 0.90, most notably for HER2-enriched and triple-negative subtypes
HEALNet — Multimodal fusion for heterogeneous biomedical data, NeurIPS 2024	Cross-attention fusion designed for heterogeneous, incomplete biomedical modalities
Multimodal multi-instance evidence fusion neural networks for cancer survival prediction	*Sci Rep* 15(1):10470 (2025) — evidence-based fusion with uncertainty, directly relevant to modality dropout
Vanguri et al., *Nature Cancer* 3(10):1151–1164 (2022)	Multimodal integration of radiology, pathology and genomics to predict response to PD-(L)1 blockade in NSCLC — the reference example of cross-scale clinical multimodal fusion
Liu et al., *Nature Communications* 16:3510 (2025)	Multimodal fusion of radio-pathology and proteogenomics identifying integrated glioma subtypes with prognostic and therapeutic implications
Subramanian, Syeda-Mahmood & Do, ISBI 2021	Sparse CCA fusion for breast cancer survival prediction — classical baseline
Tri-fusion / modality-dropout literature	Documents that **modality-dropout and VAE-based imputation** have been proposed for incomplete modalities, but uncertainty propagation and unstable convergence in deep fusion pipelines remain open problems

The gap to state on the slide: published multimodal survival models overwhelmingly assume complete modalities and report single-cohort performance. The 13–22% cross-cohort drop and the behaviour under missing modalities are acknowledged as open problems, not solved ones.

SLIDE 4 — Proposed Method & Expected Outcome

Pipeline:



Training with modality dropout: during training, randomly mask entire modalities with probability *p*. This forces the fusion layer to produce a usable representation from any available subset, rather than depending on all inputs being present.

Experimental plan:

1. **Baselines:** clinical-only Cox model; omics-only; WSI-only; naive concatenation fusion
2. **Proposed:** graph fusion with modality dropout
3. **Missing-modality stress test:** evaluate every trained model under each modality-ablation condition (omics dropped, WSI dropped, both partial). Compare graceful degradation with and without dropout training
4. **Cross-cohort test:** TCGA → CPTAC, compared against the reported 13–22% drop
5. **Risk stratification:** Kaplan–Meier curves for predicted high- vs. low-risk groups, with log-rank test

Metrics: C-index (primary), time-dependent AUC, integrated Brier score, log-rank p-value. Note on the slide that accuracy is *not* reported, because the data is censored — this alone will distinguish the proposal from B1 in an examiner's mind.

Interpretability: GNN attention over tissue regions and over pathway nodes, so risk is attributable to both morphology and biology.

Expected outcome:

- A prognostic model with C-index benchmarked against published multimodal survival work
- **A degradation curve** showing performance as a function of modality availability — the central deliverable, and something most published work does not report
- A cross-cohort C-index on CPTAC
- Kaplan–Meier separation demonstrating clinically meaningful stratification

Risks and mitigations:

- *Limited event counts in TCGA-BRCA* (relatively good prognosis, heavy censoring) → use progression-free interval alongside overall survival; report event counts explicitly; avoid over-parameterized models
- *Graph construction choices are arbitrary* → ablate over k in k-NN and over pathway database choice
- *CPTAC is small (n≈134)* → report confidence intervals; treat as a directional generalization signal, not a definitive benchmark

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How the Four Combine into One Team Project

Proposed team title: *Multimodal AI Decision Support Across the Breast Cancer Care Pathway — from Molecular Diagnosis to Treatment Response and Outcome*

Stage of care	Clinical question	Proposal	Modality	Metric
Diagnosis	What kind of cancer is this?	**B1**	Pathology + Genomics	AUC
Treatment planning	Will it respond to chemotherapy?	**A1**	MRI + Clinical	AUC
Treatment assessment	Did it respond?	**A2**	MRI (longitudinal)	Dice → Sensitivity
Outcome	What is the long-term risk?	**B2**	Pathology + Genomics (graph)	C-index

Shared infrastructure the team builds jointly (this is what makes it one project rather than four stapled together):

1. A unified data-loading layer over TCIA / GDC / Zenodo sources
2. A common fusion backbone (cross-attention module reused across all four)
3. A single evaluation harness with consistent patient-level splitting, bootstrapped confidence intervals, and standardized reporting
4. A shared interpretability module (Grad-CAM for volumes, attention rollout for MIL and GNN)

Suggested ownership: given a multi-omics and graph-modelling background, **B2** is the natural fit, with **A2** as the second choice.

Two practical cautions:

- Do **not** let the four decks converge in structure. Same disease is expected and fine; four identically formatted slide sets with the same figures will read as coordinated rather than independent. Vary the opening hook, the figures, and the citation set.
- Each proposal must stand alone. Grouping happens *after* evaluation, so no slide should reference the other three.

Master Reference List

Datasets — Imaging (Base A)

Resource	Link
BreastDCEDL (*Sci Data* 2026), 2,070 patients	https://www.nature.com/articles/s41597-026-06589-6
BreastDCEDL — Zenodo main record	https://zenodo.org/records/15627233
BreastDCEDL_ISPY2 — Zenodo (n=982)	https://zenodo.org/records/17578255
BreastDCEDL — GitHub	https://github.com/naomifridman/BreastDCEDL
BreastDCEDL_ISPY2 demo notebook	https://github.com/naomifridman/BreastDCEDL/blob/main/ISPY2/BrestDCEDL_ISPY2_demo.ipynb
MAMA-MIA (*Sci Data* 2025), 1,506 expert-segmented cases	https://www.nature.com/articles/s41597-025-04707-4
I-SPY2 / ACRIN 6698 — TCIA	https://doi.org/10.7937/TCIA.D8Z0-9T85
I-SPY2 — TCIA collection wiki	https://wiki.cancerimagingarchive.net/pages/viewpage.action?pageId=70230072
ACRIN 6698 / I-SPY2 Breast DWI	https://doi.org/10.7937/TCIA.KK02-6D95
I-SPY1 / ACRIN 6657 — TCIA wiki	https://wiki.cancerimagingarchive.net/pages/viewpage.action?pageId=20643859
Duke Breast Cancer MRI — TCIA	https://doi.org/10.7937/TCIA.e3sv-re93
TCIA portal	https://www.cancerimagingarchive.net

Datasets — Pathology / Omics (Base B)

Resource	Link
TCGA-BRCA — GDC portal	https://portal.gdc.cancer.gov/projects/TCGA-BRCA
TCGA-BRCA imaging collection	Lingle et al., The Cancer Imaging Archive, 2016 — via https://www.cancerimagingarchive.net
CPTAC-BRCA	via GDC / CPTAC Data Portal

Key Papers — Chemo Response (A1 / A2)

- Attention-based multimodal deep learning for interpretable and generalizable prediction of pCR in breast cancer — https://pubmed.ncbi.nlm.nih.gov/40640789/ *(I-SPY2 n=660, 217 pCR = 32.88%; I-SPY1 n=114, 29 pCR = 25.44%; AUC 0.73±0.04 internal / 0.71±0.02 external; MRI-only 0.68/0.70; clinical-only 0.66/0.71)*
- Two-stage dual-task learning strategy for early pCR prediction using DCE-MRI — https://pubmed.ncbi.nlm.nih.gov/40639409/ and https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12272044/ *(I-SPY2, T0/T1/T2; AUROC 0.799 → 0.820)*
- Deep learning prediction of pCR, RCB and PFS in breast cancer patients — https://journals.plos.org/plosone/article?id=10.1371%2Fjournal.pone.0280148 Code: https://github.com/HongyiDuanmu26/Prediction-of-pCR-with-Integrative-Deep-Learning *(I-SPY1 Level 3, 221 patients → 155 after exclusions, 9 institutions, 4 timepoints)*
- Deep learning models based on pretreatment MRI and clinicopathological data for TNBC — https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11940201/ *(282 patients; AUC 0.76 internal, 0.72 on external I-SPY2)*
- Prediction of early NAC response through DL-based pharmacokinetic quantification of DCE-MRI — https://pmc.ncbi.nlm.nih.gov/articles/PMC12464716/ *(ACRIN 6698 / BMMR2 challenge datasets A1/A2, plus I-SPY2 and I-SPY1)*
- A novel CNN algorithm for pCR prediction using an I-SPY TRIAL breast MRI database — https://www.sciencedirect.com/science/article/abs/pii/S0730725X20302733 *(improved InceptionV3; precision 0.83, recall 0.77, F1 0.80, accuracy 0.81, AUC 0.81)*
- MNJ-supplied reference — https://doi.org/10.4238/n53tzp26

Key Papers — Multimodal Pathology / Omics (B1 / B2)

- M3FusionNet: Cross-cohort multimodal prediction of breast cancer biomarkers — https://pubmed.ncbi.nlm.nih.gov/42407417/ *(TCGA-BRCA n≈1036, CPTAC-BRCA n≈134; in-distribution AUC 0.99 ER / 0.96 PR / 0.98 HER2 / 0.96 PAM50 macro, overall 0.97; cross-dataset drop 13–22%, E1 overall 0.82; combined E3 overall 0.90; Macenko + ComBat)*
- Selective multimodal deep learning for reliable breast cancer subtype classification from histopathology and genomic data — https://pubmed.ncbi.nlm.nih.gov/41671586/ *(TCGA-BRCA, 924 patients with PAM50; 94.93% accuracy; multimodal inference for only 38.2% of samples; 3.12× speedup)*
- MM-SurvNet: Deep learning-based survival risk stratification in breast cancer through multimodal data fusion — https://arxiv.org/pdf/2402.11788

- Cross-Attention Multimodal Fusion for Breast Cancer Diagnosis: Integrating Mammography and Clinical Data with Explainability — https://arxiv.org/pdf/2508.16000 *(contains a comparison table of 2023–2025 multimodal breast cancer models: DeepClinMed-PGM, CVFBJTL-BCD, MMFF-Net, MFDNN, LightweightUNet)*
- Explainable multi-modal deep learning for transparent cancer diagnosis — https://www.frontiersin.org/journals/artificial-intelligence/articles/10.3389/frai.2026.1767612/full *(CBIS-DDSM + Duke Breast MRI + TCGA BRCA/LUAD/GBM; 3,842 images, 2,917 patients; Grad-CAM++ and SHAP)*
- Multimodal Fusion of Imaging and Multi-omics Data for Enhanced Breast Cancer Detection — https://link.springer.com/chapter/10.1007/978-3-032-18474-0_36 *(8 modalities on TCGA-BRCA: MRI, dual-resolution histopathology 20×/40×, RNA-Seq, CNV, methylation, miRNA, proteomics)*
- Hemker, Simidjievski & Jamnik. HEALNet: Multimodal fusion for heterogeneous biomedical data. NeurIPS 2024.
- Luo et al. Multimodal multi-instance evidence fusion neural networks for cancer survival prediction. *Sci Rep* 15(1):10470 (2025). https://doi.org/10.1038/s41598-025-93770-3
- Vanguri et al. Multimodal integration of radiology, pathology and genomics for prediction of response to PD-(L)1 blockade in NSCLC. *Nature Cancer* 3(10):1151–1164 (2022).
- Liu et al. Multimodal fusion of radio-pathology and proteogenomics identify integrated glioma subtypes. *Nature Communications* 16:3510 (2025).
- Subramanian, Syeda-Mahmood & Do. Multimodal fusion using sparse CCA for breast cancer survival prediction. ISBI 2021, pp. 1429–1432.

Tooling / Methods References

- Macenko et al. A method for normalizing histology slides for quantitative analysis. ISBI 2009, pp. 1107–1110.
- Bankhead et al. QuPath: Open source software for digital pathology image analysis. *Sci Rep* 7:16878 (2017).
- Li & Dewey. RSEM: Accurate transcript quantification from RNA-Seq data. *BMC Bioinformatics* 12:323 (2011).
- Clark et al. The Cancer Imaging Archive (TCIA): Maintaining and Operating a Public Information Repository. *J Digit Imaging* (2013). https://doi.org/10.1007/s10278-013-9622-7
- Parker et al. PAM50 intrinsic subtype classifier (referenced in MM-SurvNet).
- nnU-Net — baseline segmentation framework; pretrained weights supplied with MAMA-MIA.

Pre-Submission Checklist

☐ Confirm with course coordinators whether the LVPEI-style "sample data" claim also applies to MNJ/Kamineni (i.e. what will actually be provided, and when)

☐ **A2 only:** verify how many post-chemotherapy (T4) timepoints carry segmentation masks in the I-SPY1/I-SPY2 releases; if sparse, pivot to longitudinal T0→T1→T2 change modelling

☐ Download at least one dataset and open it before pitching — being able to say "I have already loaded the data" is worth more than any slide

☐ Confirm no two of the four decks share figures or layout

☐ Each deck must stand alone — no cross-references to the other three proposals

- [ ] Practise the 10-minute timing: roughly 2 min problem, 2 min data, 2.5 min SOTA, 3.5 min method and outcome